

Companion LV: The CMB-Lensing $\times$ Galaxy Cross-Correlation in the Relaxation-Driven Cyclic Cosmology

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Abstract

The cross-correlation between CMB lensing and galaxy density provides a powerful probe of the relationship between matter and the gravitational potential. In the standard cosmological model, this correlation is determined by the matter power spectrum, the galaxy bias, and the lensing kernel. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the lensing kernel, the matter distribution, and the growth rate in a characteristic way. This Companion develops a continuous description of the CMB-lensing $\times$ galaxy cross-correlation in RDCC, deriving how the relaxation scale influences the coupling between matter and lensing, the coherence of large-scale modes, and the observational signatures in galaxy surveys and CMB experiments. The resulting framework provides a distinctive observational signature of the RDCC gravitational field.

1 Introduction

The cross-correlation between CMB lensing and galaxy density probes the connection between the gravitational potential and the matter distribution. CMB lensing traces the integrated potential along the line of sight, while galaxy surveys trace the matter density at specific redshifts. Their cross-correlation therefore encodes the relationship between matter and the gravitational field.

In the standard cosmological model, this correlation is determined by the matter power spectrum, the galaxy bias, and the lensing kernel. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the lensing kernel, the matter distribution, and the growth rate in a scale-dependent manner, producing a distinctive signature in the CMB-lensing $\times$ galaxy cross-correlation.

2 The RDCC Lensing Kernel

The CMB lensing potential is given by

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi \frac{\chi_* - \chi}{\chi_* \chi} [\Phi + \Psi](\chi, \hat{\mathbf{n}}). \quad (1)$$

In RDCC, the metric potentials evolve as

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right], \quad (2)$$

$$\Psi_{\text{RDCC}}(k, a) = \Psi(k, a) \left[1 - \chi_\Psi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (3)$$

The RDCC lensing kernel becomes

$$W_{\phi,\text{RDCC}}(\chi) = W_{\phi}(\chi) \left[1 - \chi_{\phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (4)$$

3 The RDCC Galaxy Density Field

The galaxy density contrast is given by

$$\delta_g(k, z) = b(z) \delta_m(k, z), \quad (5)$$

where $b(z)$ is the galaxy bias.

In RDCC, the matter density evolves as

$$\delta_{m,\text{RDCC}}(k, z) = \delta_m(k, z) \left[1 - \chi_{\delta} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (6)$$

The galaxy density becomes

$$\delta_{g,\text{RDCC}}(k, z) = b(z) \delta_m(k, z) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (7)$$

4 The RDCC Lensing $\times$ Galaxy Cross-Correlation

The cross-correlation is defined as

$$C_L^{\phi g} = \int d\chi \frac{W_{\phi}(\chi) W_g(\chi)}{\chi^2} P_m \left(k = \frac{L}{\chi}, z(\chi) \right). \quad (8)$$

In RDCC, this becomes

$$C_L^{\phi g, \text{RDCC}} = C_L^{\phi g} \left[1 - \chi_{\phi g} \left(\frac{L}{L_{\text{rel}}} \right)^{\alpha} \right]. \quad (9)$$

This modification reflects the scale-dependent evolution of the gravitational potential and the matter distribution.

5 Large-Scale Coherence and RDCC Signatures

The relaxation mechanism enhances the coherence of large-scale modes. The CMB-lensing $\times$ galaxy cross-correlation therefore exhibits:

- enhanced power at large angular scales,
- suppressed power at small scales,
- a characteristic turnover near L_{rel} ,
- a modified redshift dependence of the correlation amplitude.

These signatures provide a direct observational probe of the RDCC gravitational field.

6 Conclusion

The CMB-lensing $\times$ galaxy cross-correlation in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the matter distribution. The relaxation mechanism modifies the lensing kernel, the matter density field, and the growth rate in a scale-dependent manner, producing distinctive signatures in the cross-correlation spectrum. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of CMB–LSS coupling, matter evolution, and the interplay between light and gravity in RDCC.

References

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